

PXT992 Micro Project Diary

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1 Week 5: 11 December 2023 – 15 December 2023

1.1 Week 5 objectives

1. Read as much research papers as I can;
2. Complete the diary on time;
3. Write a draft of the microproject plan;

1.2 Monday 11 December 2023

I found a paper, "Explorations in Hubble Space: A Quantitative Tuning Fork" by ROBERTO G. ABRAHAM and MICHAEL R. MERRIFIELD.[1], talks about a way, developed to quantify two-parameters that describes galactic structures that maps closely to the Hubble's original tuning fork. The classification criteria that Hubble used were basic, but the complexity of the galaxy morphology really does not are not covered in the classification. This paper tries to provide a better way or atleast tries to explain a way that we can classify galaxies better. The modern technology has grown to an extent where just the visual classification of the galaxies are not practical anymore. To overcome this difficulty, multiple techniques are employed fro the classification:

- Visual classification using coarse bins that neglect the visibility of spiral structure.[1] [2];
- Classification based on surface-brightness profile fits to analytical models.[1][3];
- More general automated schemes based on various combinations of image concentration and surface brightness, image concentration and asymmetry, and image concentration, asymmetry, surface brightness, and spectral class.[4][5][6][7]

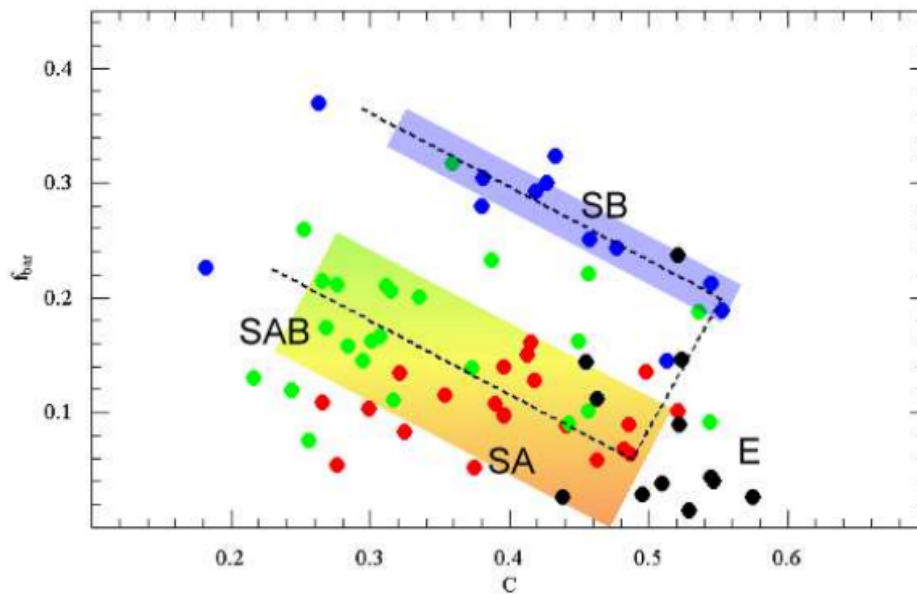


Figure 1.1: Distribution of galaxies. Unbarred galaxies are shown in red, weakly barred galaxies are shown in green, strongly barred systems are shown in blue, and elliptical galaxies are shown in black. The dashed lines indicates a schematic tuning fork. Shaded regions in the figure correspond to 1σ rectangular contours delineating the two lines of the quantitative tuning fork.[1]

1.3 Tuesday 12 December 2023

No work done.

1.4 Wednesday 13 December 2023

No work done.

1.5 Thursday 14 December 2023

Microproject meeting at 2:30PM in McKenzie house. Talked about how the microproject plan is written, What to and what not to include in the same.

1.6 Friday 15 December 2023

Diary submission due.

1.7 Operational summary for week 5

- Read few papers this week which helped me to start writing the Literature review.
- Submitted the diary.
- Micro-project plan draft not ready.

1.8 Reflective statement for week 5

- Did not start writing the microproject plan. Keeping the diary updated everyday would have helped me from not rushing through editing the diary properly.
- Should have started using Latex but I did not want to step into an unknown thing right now. Hesitant because my friends told me, latex has a bit of learning curve but it is better than word.

2 Week 6: 18 December 2023 – 20 December 2023

2.1 Week 6 objectives

- Complete the Literature review.
- Study for the exams.

2.2 Monday 18 December 2023

No work done

2.3 Tuesday 19 December 2023

Writing the microproject plan.

2.4 Wednesday 20 December 2023

Microproject plan due (Submitted).

Among the requested galaxy data from the LCO archive, the data for one of the galaxy (NGC891) was available when I checked today.

2.5 Operational summary for week 6

- Did not start the draft of Literature review.
- Microproject plan and diary have been submitted.

2.6 Reflective statement for week 6

The plan could have been much better. If I had researched properly about the steps involved in the actual analysis of the data, maybe I could have had a better plan. But I do not think the one I have right now is too shabby.

3 Week 7: 08 January 2024 – 12 January 2024

3.1 Week 7 objectives

- Keep checking the LCO archive for the requested data.
- Finish the literature review and reflective statement

3.2 Monday 08 January 2024

I looked at the LCO website for the requested data. There were data of 6 galaxies available out of the 6 galaxies requested. (NGC3982, NGC7773, NGC2865, NGC891, NGC4013, NGC4565)

3.3 Tuesday 09 January 2024

Revision of Experimental Gravitation 1

3.4 Wednesday 10 January 2024

Revision of cosmology and General relativity

3.5 Thursday 11 January 2024

Finishing up literature review. Draft of the reflective statement.

3.6 Friday 12 January 2024

Finish the reflective statement.

Reflective statement and Literature review submissions.

3.7 Operational summary for week 7

- Both the objectives set for this week have been met.
- Writing up the reflective statement was in a bit of hurry.

3.8 Reflective statement for week 7

- Maybe, I could have started the reflective statement sooner.
- More papers could have been read this week.
- I have got the data that I am going to analyse which is a progress in the project.
- Busy and stressed with the exams. Could not focus much on the micro-project part of things. I chose to give importance to the exams and preparing for it.

4 Week 8: 29 January 2024 – 02 February 2024

4.1 Week 8 objectives

- Read through few articles on the photometry and software packages.
- Try to figure out which software to use and get it working.
- Start the analysis of the data.

4.2 Monday 29 January 2024

No work done.

4.3 Tuesday 30 January 2024

No work done.

4.4 Wednesday 31 January 2024

No work done.

4.5 Thursday 01 February 2024

No work done.

4.6 Friday 02 February 2024

Micro-project meeting started around 12:00PM at McKenzie House 4th floor. This is the first meeting after the Christmas break. We, galaxy group, have got the data of all 6 galaxies that we are going to analyse. I still have problem with the software. APT does not work, says something about JAVA being not in the computer. Wondering whether I should be using a different software for time being.

The galaxy group i.e. Liam, Shyam and me, three of us got into a discord group call to figure out the surface brightness profile. Since Liam had experience with a similar project in his undergrad, the following is how he taught us to take the radial aperture measurement on the galaxies.

1. Open the JS9 web page on a browser <https://js9.si.edu/>
2. Click on 'File' and then on 'Open local'.
3. Find the folder where your Fits files are stored, select one and click open.
4. The data will be opened on the page as a 2D array.
5. Now click on the annulus (a circle) button towards the upper left side on the bar.
6. You can see a green circle on the image now. Double click on the circle to open a 'Edit region' menu.
7. On the 'Edit' option, enter numbers from 1 to 10 (or how many ever radii you want) which is then converted into arcsec by the software itself.
8. You'll get concentric circles, you can adjust it to fit the object in the image.
9. Go to the bar on the top and click 'Analysis' and then 'Counts in Regions' in the menu that shows up.
10. This will give you a list of surface brightness and surface errors along with other things.
11. Copy and paste this into an excel sheet which then can be imported into the python program to draw a surface brightness profile.
12. You can also use the plot function in the Microsoft excel to produce a graph, which is a surface brightness profile.

4.7 Operational summary for week 8

- We did start the analysis process.
- Ended up not deciding on a software that has to be installed on a computer. Instead, started using JS9[8] software which is hosted on an online server.
- Started the analysis process.

4.8 Reflective statement for week 8

- Could have spent more time figuring out the analysis part of it.
- Coming out of the holidays mood and the galaxy group meeting on discord helped to kick start the interest in doing more on the project.

5 Week 9: 05 February 2024 – 09 February 2024

5.1 Week 9 objectives

- Find a way to write the codes to do the photometry.
- Plotting apertures using python.
- Get the list of flux values.

5.2 Monday 05 February 2024

I have found this playlist on YouTube by a channel called Desi Astro. He is an Indian explaining how to write the photometry codes in Python using Photutils[9]. He does not speak english very well but his codes work fine. I started just skimming through the videos before starting to type that up. It looks trustworthy. I have started with the code to install Photutils package in the colab file. The following are the packages needed to work through this analysis or photometric analysis.

NOTE: All the codes used is from the [youtube playlist](#) and I had to rewrite to fit my data and situation. I have written lines of code from scratch as well using pre-defined functions in Photutils [9].

Listing 1: Installing Packages(taken from Desi Astro YT)

```
!pip install photutils
!pip install astropy
!pip install ipynb
```

This will take some time to run and to install the packages.

Listing 2: Importing Packages(taken from Desi Astro YT)

```
#importing libraries
from pylab import *
from astropy.io import fits
import astropy as ap
import matplotlib.pyplot as plt
from matplotlib.colors import LogNorm
import numpy as np
import photutils
from photutils.aperture import CircularAperture
```

Here, 'pylab' is a package that has all the mathematical and usual functions that helps to write a piece of code. 'Astropy' is a community core package which is made for astrophysics and is used majorly throughout the field. 'matplotlib' is mainly used to plot graphs (complex graphs). 'numpy' is a package used for array and matrix creation and manipulation. 'photutils' is another package associated with astropy for astrophysics, which gives tools to detect and perform photometry on astronomical objects.

5.3 Tuesday 06 February 2024

No work done.

5.4 Wednesday 07 February 2024

I have chosen to use python to do everything, from importing the fits file to plot the surface brightness and for all the photometric analysis.

Listing 3: Importing the FITS file(taken Desi Astro YT and modified)

```
file1 = fits.open('/content/ogg2m001-ep02-20231228-0071-e91.fits.gz')
NGC7773=file1[1].data
file2 = fits.open('/content/ogg2m001-ep02-20231228-0886-e91.fits.gz')
NGC3982=file2[1].data
file3 = fits.open('/content/ogg2m001-ep02-20231231-0660-e91.fits.gz')
```


number that should be read or accessed. In this case the actual values we need to show the image is stored in the first row of the variable. This can be seen if you observe the output from the python code mentioned above in Listing 4.

After the required data is stored in a variable (here for example NGC7773 (look at Listing 3) (Naming the variable with the name of the galaxy will make the identification of the image easier.

Now plotting or producing the image of the galaxy from the data, the flux values are plotted as a 2D array using matplotlib package. The following code was used to plot the 2D array:

Listing 5: Output the data in the FITS file(Written by me)

```
#plot the fits file data
%matplotlib ipympl
plt.figure()
plt.imshow(NGC7773, norm=LogNorm(), origin='lower', cmap='jet')
plt.title('NGC7773')
plt.xlabel('Pixels')
plt.ylabel('Pixels')
plt.colorbar()
plt.show()
```

'matplotlib ipympl' is used to make the plot interactive. Used to zoom into the plot. The plot consists of pixels in X and Y axis, the values that are stored in the variable, pixel values. The image below is the output of the code in Listing 5.

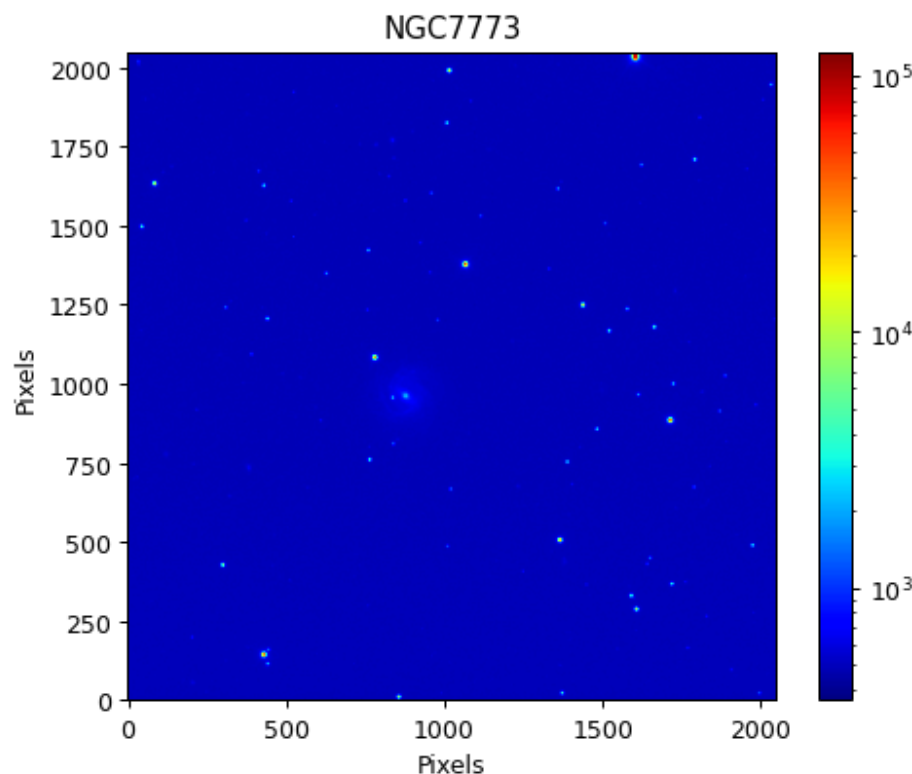


Figure 5.1: Original data of the galaxy NGC7773

5.6 Friday 09 February 2024

Micro-project meeting started around 11 o'clock in the morning at the usual place (McKenzie House). I had to mention what my progress was regarding the results of the project. I should have got the surface brightness profile but I wanted to write a line of code to get the highest flux value instead of manually zooming in and guessing the center of the galaxy. Highest flux value because, the center of the galaxy has the highest luminosity. I am struggling with the interactive plot behaving unpredictably when the codes are run.

5.7 Operational summary for week 9

- Figured out how to import fits into python environment.
- Did not plot the apertures using python package.
- Did not reach the stage where I can plot the surface brightness.

5.8 Reflective statement for week 9

- I have spent satisfying time on the project, figuring things out myself.
- Plotting the image made me happy.

6 Week 10: 12 February 2024 – 16 February 2024

6.1 Week 10 objectives

- Get the surface brightness profile.
- Try to figure out the code that gets the highest flux value.

6.2 Monday 12 February 2024

I wrote the code to plot the 2D array image of the other 2 face on galaxies. I found that NGC7773 and NGC2865, two face on galaxies had astronomical objects, stars, near them, in our field of vision.

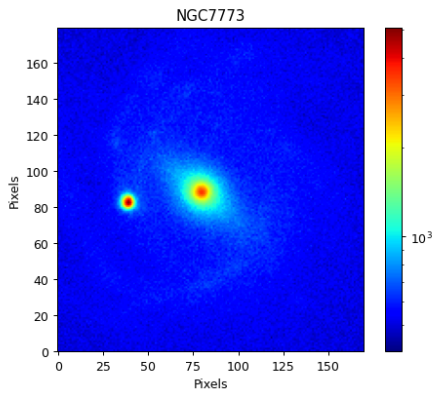
As seen in figure 5.1 the galaxy is too small to see and draw the apertures. In order to crop the image the following code was used:

Listing 6: Crop the image (Written by me)

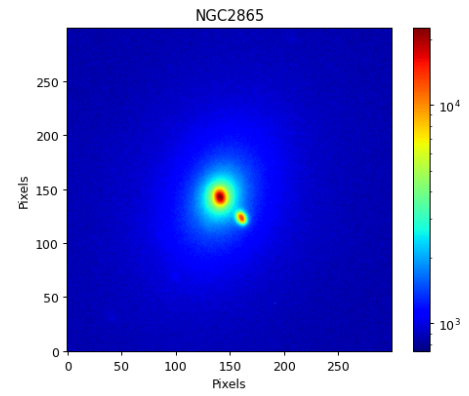
```
NGC7773crop=NGC7773[870:1050,800:970]
plt.figure()
plt.imshow(NGC7773crop, norm=LogNorm(), origin='lower', cmap='jet')
plt.title('NGC7773')
plt.xlabel('Pixels')
plt.ylabel('Pixels')
plt.colorbar()
plt.show()
```

It is not possible to draw the aperture with just the original image, cropping in i.e. zooming in to the image gives more better picture of the object.

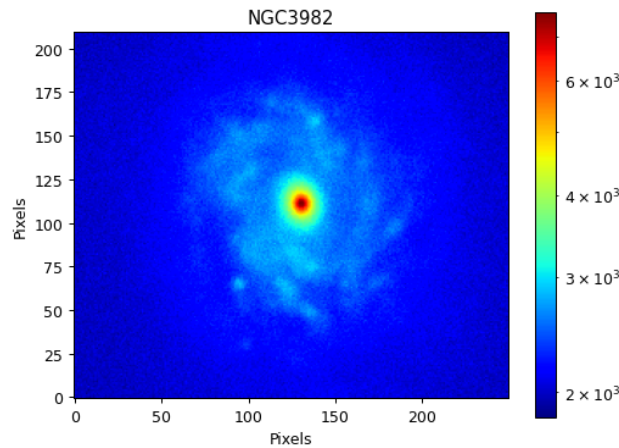
The first line of the code snippet crops the image into the part I want. [x1 : x2 , y1 : y2] is the general syntax used to define the range of X and Y axis values to be plotted. Figuring out the values of x1, x2, y1, y2 is a bit time consuming and only way to figure it out is enter the estimate the co-ordinates by trial and error method. Here, 'cmap' is where the type of filter is mentioned to be applied on the image while it is output.



(a) NGC7773



(b) NGC2865



(c) NGC3982

Figure 6.1: Cropped images of the face-on galaxies

6.3 Tuesday 13 February 2024

No work done.

6.4 Wednesday 14 February 2024

I have got cropped images of 3 face-on galaxies. For now I am using these images to do the photometric analysis. I will look into the other 3 edge-on galaxies later since performing analysis on them with the time frame of this project is not feasible.

Writing the code to get the highest value of flux in the image:

Listing 7: Highest flux value (Written by me)

```
X3928,Y3982= np.unravel_index(NGC3982crop.ravel().argmax(), NGC3982crop.shape)
print("Co-ordinates of highest flux value in
cropped image of NGC3982:",X3928,Y3982)
```

```
X2865,Y2865= np.unravel_index(NGC2865crop.ravel().argmax(), NGC2865crop.shape)
print("Co-ordinates of highest flux value in
cropped image of NGC2865:",X2865,Y2865)
```

```
X7773,Y7773= np.unravel_index(NGC7773crop.ravel().argmax(), NGC7773crop.shape)
print("Co-ordinates of highest flux value in
cropped image of NGC7773:",X7773,Y7773)
```

Here,'unravel index' is a numpy package function which converts an array of flat indices into tuple of coordinate arrays. The .ravel() function creates flattened array of the co-ordinates and argmax() function returns the

highest value in the array. I have cropped the image because the unravel function converts all the pixels included in the image into an array.

Listing 8: Output of the highest flux value

```
Co-ordinates of highest flux value in cropped image of NGC3982: 111 129
Co-ordinates of highest flux value in cropped image of NGC2865: 142 141
Co-ordinates of highest flux value in cropped image of NGC7773: 82 39
```

If you look at the pixel numbers obtained for NGC7773 galaxy, 82 and 39, are not of the center of galaxy. It is of the star that is in the field of view. Either I have to find a way to subtract this object from the image or reduce the luminosity of the star from the measurements I take of the galaxy.

6.5 Thursday 15 February 2024

I have the co-ordinate values that I can draw the radial aperture on. The following code uses python to plot multiple radial apertures keeping the co-ordinate value I have got for each images from the unravel function in Listing 8. For now I am choosing to do the photometric analysis only on the galaxy NGC3982 because it is clear and there are not other objects obstructing it.

Listing 9: Drawing apertures(taken from Desi Astro YT and modified)

```
#Center of NGC3982 in pixels 111,129
#Drawing the aperture

%matplotlib ipynpl
flux_list=[]
area_list=[]
plt.figure(figsize=(5,5))
from photutils.aperture import CircularAperture, ApertureStats
plt.imshow(NGC3982crop,norm=LogNorm(),origin='lower',cmap='jet')
rad=np.array(linspace(1,55,100))
for i in range(len(rad)):
    aper3=CircularAperture([129,111],rad[i])
    aper3.plot(color='red',lw=2)

#find flux in the corresponding aperture
#phot_table=aperture_photometry(NGC3982crop, aper3)
#print(i, phot_table)
aperstats= ApertureStats(NGC3982crop, aper3)
area_list.append(aperstats.sum_aper_area)
flux_list.append(aperstats.sum)
print(aperstats.sum,aperstats.sum_aper_area)
```

Here, the program calculates the total flux inside one circle and returns the value [9]. Drawing multiple concentric circles will return the total fluxes in each of the circle.

Taking a look at the code:

1. I have defined a list (flux list=[]) to store all the flux values later on.
2. Importing functions 'CircularAperture' and 'ApertureStats' from photutils to draw the apertures and calculate the fluxes respectively.
3. rad=np.array(linspace(1,55,100)) defines 100 values from 1 to 55 equally spaced.
4. I have used a for loop to loop through the radius values.
5. CircularAperture([129,111],rad[i]) draws circles in each of the radius value starting from 129,111 co-ordinate as its center.
6. ApertureStats(NGC3982crop, aper3) measures the flux values inside the areas stored in aper3 on the image NGC3982crop.

7. Values are then appended onto the list using append function.

The list of flux values obtained: [23128.22750940397, 54427.56283603193, 97248.04231243004, 151192.95589368057, 213185.0303645463, 283816.06166498025, 360501.9088244212, 443433.7323004563, 531288.5894990459, 623755.923742152, 720482.037395118, 821112.4640611995, 926229.9294766893, 1034154.7935980553, 1146261.7497611826, 1261322.7060830886, 1379846.2013792144, 1501029.964971167, 1626081.8773809269, 1754142.233015655, 1885632.526191834, 2020446.2867456079, 2158204.8766125296, 2299231.2213501874, 2442926.214896825, 2590168.734660864, 2740678.337822733, 2895184.356030683, 3052704.827569405, 3213860.4886554875, 3378333.0250072544, 3546378.830793565, 3717779.61034473, 3892729.8995335563, 4071751.207745347, 4254608.30910312, 4441796.439324667, 4633435.527188744, 4829411.392936575, 5029262.614319408, 5233050.6572301835, 5441496.1421116935, 5654828.832922474, 5872709.098169081, 6095551.584666265, 6322848.204376401, 6555035.017775075, 6792291.793992067, 7034251.168259602, 7280955.167611577, 7532424.855930312, 7788999.024288732, 8050413.152909187, 8316739.3889560755, 8587901.177196369, 8863416.376440067, 9143427.862689488, 9428304.543811573, 9718074.281388264, 10012666.210417878, 10311835.793894388, 10616040.873070324, 10925678.982499495, 11240403.537306176, 11560127.196598407, 11884900.055015199, 12214617.412372459, 12548874.256542657, 12887236.194285316, 13230207.760856248, 13577438.10742496, 13929604.28644864, 14285828.541388495, 14645744.057220891, 15010183.033479754, 15379439.029299054, 15752545.51396447, 16128782.018128943, 16509092.997042835, 16892698.760883667, 17280239.49736799, 17671253.873990048, 18065529.79743042, 18462489.276529, 18862836.00707893, 19267382.647970185, 19675522.963689916, 20086778.824887678, 20500907.75868218, 20918519.558715187, 21340270.890713, 21765247.0443948, 22193458.379770603, 22624407.253340475, 23059182.3869651, 23498233.82063038, 23940567.716466106, 24386243.179308906, 24835328.77895645, 25287679.209141538]

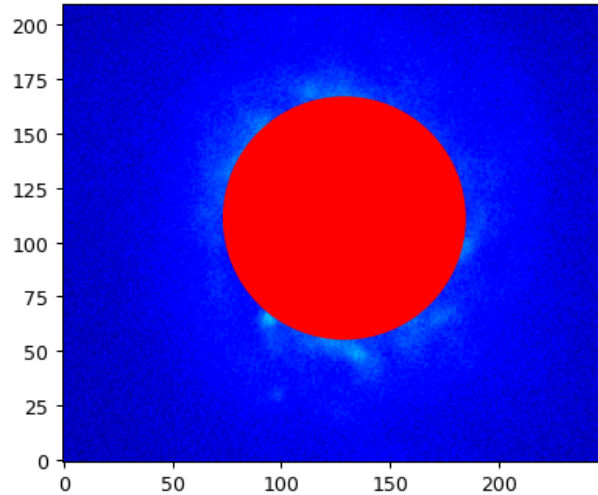


Figure 6.2: Apertures on the galaxy NGC3982

6.6 Friday 16 February 2024

Plotting the graph using scatter function in matplotlib package.

Listing 10: Plot the surface brightness (Not right)(taken from Desi Astro YT)

```
plt.figure()
plt.scatter(rad,(flux_list), marker='o', s=10)
plt.plot(rad,(flux_list))
plt.title('Curve-growth')
plt.xlabel('Radius[Apperture]')
plt.ylabel('Flux-value')
```



```
plt.show()
```

The figure is what I am getting after I run the code:

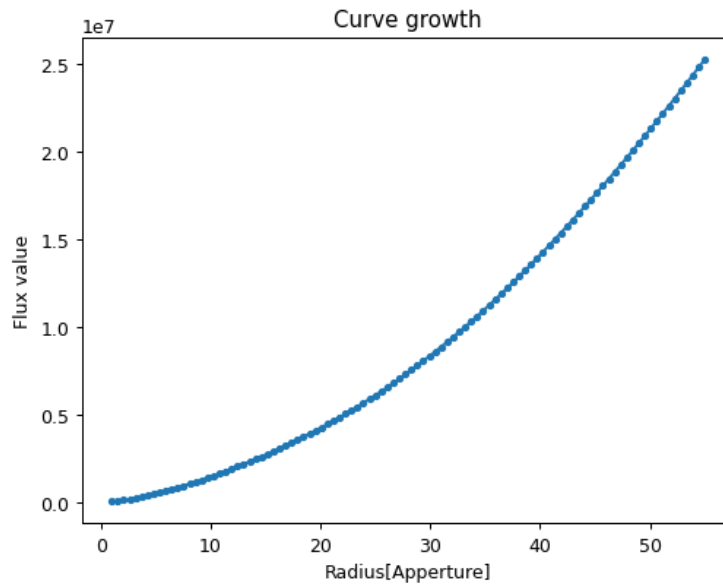


Figure 6.3: Curve growth

I then plotted the flux values in log scale by just adding the log function.

Listing 11: Plot curve growth with log on y-axis (Written by me)

```
plt.scatter(rad,log(flux_list), marker='o', s=10)  
plt.plot(rad,log(flux_list))
```

Which gave me a graph that looked like Fig 6.4

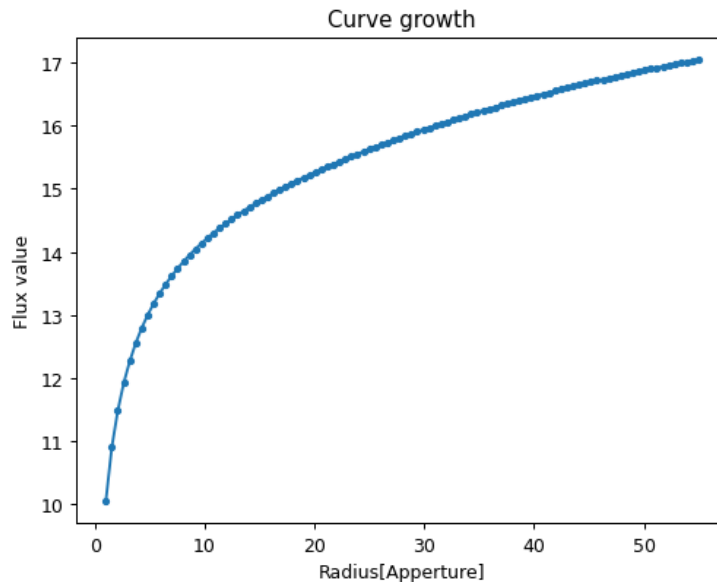


Figure 6.4: Curve growth

The micro-project meeting was held as usual. I am still trying to get the surface brightness right. Paul had instructed everyone in the group to start writing the draft of the research paper. I have not got any results for a draft. After the meeting, Dipanjan, Shyam and Liam helped me understand how to calculate the flux density with the values I have got.

6.7 Operational summary for week 10

- Figured out the code that gives the highest flux value.
- Surface brightness profile, not yet there. But I have got some form of profile.

6.8 Reflective statement for week 10

- Figuring something out myself makes me happy. I did that with the line of code that gave me highest flux value.
- Cropping the image was another hassle. I had go with trial and error method to figure out where the galaxy co-ordinates were in the image.

7 Week 11: 19 February 2024 – 23 February 2024

7.1 Week 11 objectives

- Get the surface brightness right.
- Try to fit the single sersic function.

7.2 Monday 19 February 2024

The main difference between using a well known photometry software and python to analyse an astronomical image is in python it has to be programmed to perform any process. With software's, background subtraction is automatic and the values are avoid of the background flux values. Aperture photomtery function in photutils assumes the background has been subtracted[9]. There are lines of code that can do the process mentioned here [9]. But for now I have just used the flux values that I have got.

I have converted the radius values, which is in pixels, into arc seconds by multiplying 0.27 with each values of radius and storing it in a new list. MuSCAT 2m Faulkes telescope has a pixel scale of 0.27 arcsec per pixel[10].

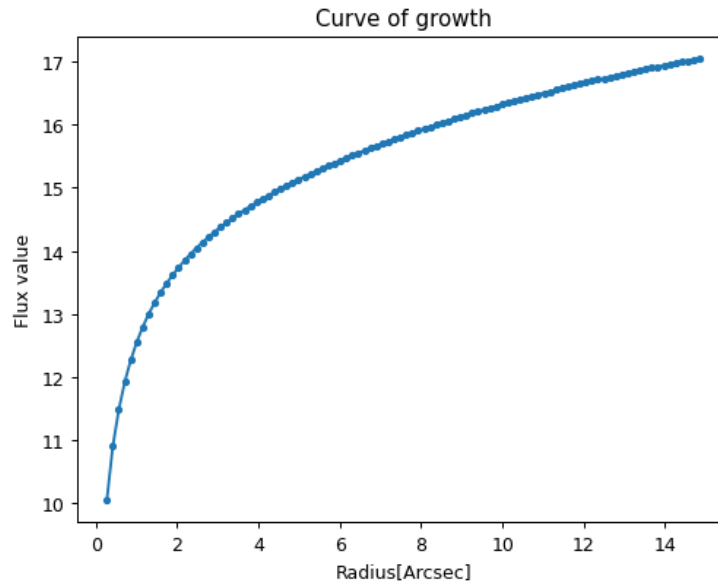


Figure 7.1: Curve growth with radius in arcseconds

7.3 Tuesday 20 February 2024

Now that I have the right values of flux and radii, I have calculated the flux densities using the equation:

$$\text{Flux density}(i+1, i) = \text{flux} / \pi(r_{i+1}^2 - r_i^2) \quad [11] \quad (7.1)$$

Here is the code I used:

Listing 12: Finding Flux density (Written by me. Modified from Desi Astro YT)

```
#section 1
radiussq=rad_arc**2
raddiff=np.diff(radiussq)
diff_r=np.append(rad_arc[0]**2,raddiff)

#section 2
flux_diff=np.diff(flux_list)
flux_diff1=append(flux_list[0],flux_diff)
flux_density=flux_diff1/(np.pi * diff_r)
print((flux_density))
```

```

#section 3
plt.figure()
plt.scatter(rad_arc , flux_density , s=15)
plt.loglog(rad_arc , ( flux_density ))
plt.title( 'Radial-Flux-Profile ' )
plt.xlabel( 'Radius[ Arcsec] ' )
plt.ylabel( 'Flux-Density ' )
plt.show()

```

Section 1 squares the radius values and finds the difference between the consecutive values. I am appending the first value of the radius stored in the radius list earlier into the final list after the squaring and the difference. I am doing this because when I use the np.diff function it does not take the difference of the first value which in turn reduces the length of the list by one.

Section 2 takes the difference between the consecutive flux values and appends the first flux value from the list earlier for the same reason as of the radius. But the equation does not have any differences taken between the fluxes, I am taking the differences because I need the flux density values in between each circle, not the entire area of each circle.

Section 3 plots these flux density values against the radius.

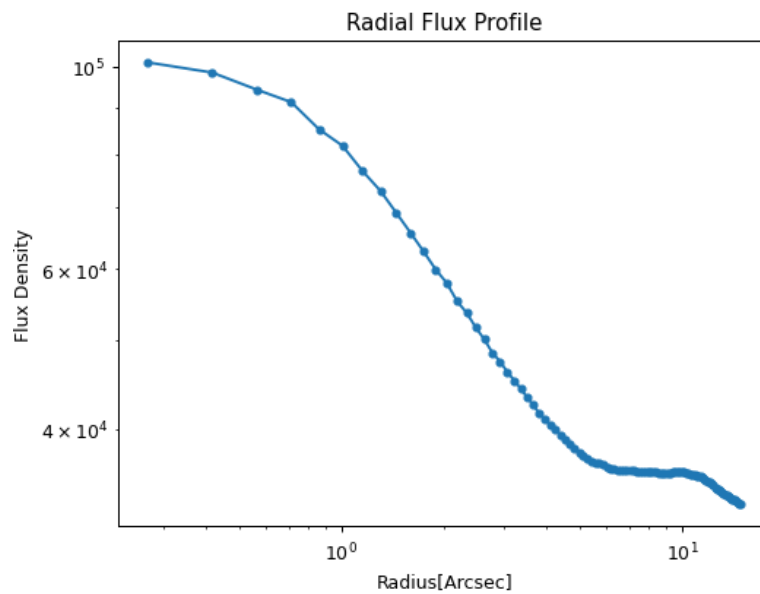


Figure 7.2: Radial Flux Profile

7.4 Wednesday 21 February 2024

No work done.

7.5 Thursday 22 February 2024

I have calculated the magnitude which is inversely proportional to the flux density.

Listing 13: Finding magnitude (Written by me)

```

#conversion of flux into magnitudes
mag=22.5-np.log10( flux_density )

```

```

#Conversion into arcsec
mag_arcsec=22.5-np.log10( flux_density / (0.27)**2 )

```

For this project there is no need of this calculation since I am trying to fit the sersic fit onto the radial flux profile.

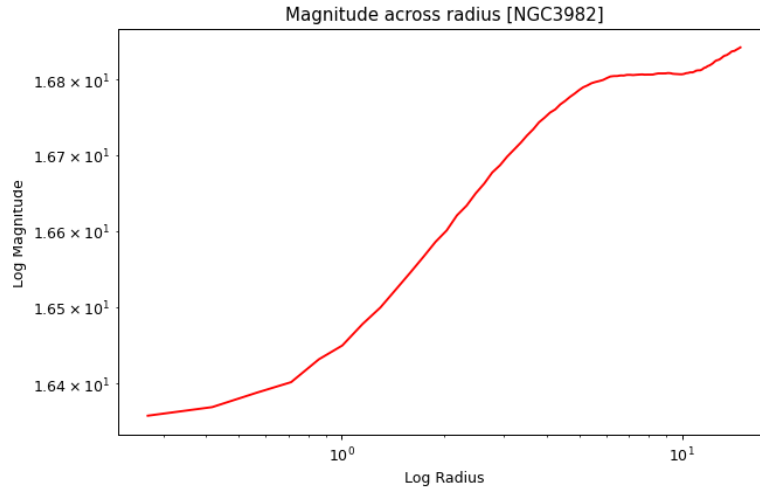


Figure 7.3: Magnitude across radii

7.6 Friday 23 February 2024

I had a glance through the piece of code of sersic fit which is uploaded on Teams by Dr.Paul, it is handy to have that code. I did not have to figure it out from the scratch. I did not clearly understand what is going on in the double sersic fit code. I chose to first plot the single sersic fit.

Listing 14: Defining single sersic funtion (taken from the provided code on Teams)

```
# Define Sersic function as a function
def sersic_single(R, i_b, r_b, n_b):
    return Sersic1D(i_b, r_b, n_b) (R)
```

Above mentioned code snippet is defining the sersic single function which returns another Sersic1D function. Sersic1D [12] function uses the the de Vaucouleurs $r^{1/4}$ model [13] to model the curve and returns the values.

Listing 15: Single Curvefit (taken from the provided code on Teams and modified)

```
optgal, covgal = curve_fit(sersic_single, rad_arc, flux_density,
    bounds=((0, rad_arc[0], 0), (1e5, rad_arc[len(rad_arc)-1], 10)), maxfev = 10000)
```

Listing 15 contains the line of code that fits the sersic model to the obtained radial flux profile i.e. surface brightness profile. **Curve fit** takes in the radius and flux value with the model that I am using (sersic single) along with the bounds to produce the close approximation of the radial flux curve. This modelling of the closest fit to the curve is done through integrating the sersic1D model that is passed in the curve fit syntax. As seen in Listing 15, I got values that are stored in two variables. 'optgal' variable contains the best fit parameters which then I pass to the sersic single function defined, producing a curve according to the model. The bounds defined here, (0,rad arc[0],0),(1e5,rad arc[len(rad arc)-1],10), the first set of values (values in the first bracket) corresponds to the first values of I, R, n respectively and the second set of values corresponds to the second values of I, R, n respectively. This input gives range of values within which the curve fit function can assume the parameters to be.

The sersic1D model has three parameters - intensity, effective radius and sersic index. I have tried to get the best values for these three parameters using 'curvefit' function in 'optimize' package of python. Next step is to plot.

Listing 16: Plot Single Sersic fit (taken from the provided code on Teams)

```
print('Intensity:-', optgal[0])
print('Radius:-', optgal[1])
print('Sersic-index:-', optgal[2])
plt.figure()
plt.scatter(rad_arc, flux_density, s=15, label='Data')
plt.loglog(rad_arc, (flux_density), label='Data')
plt.loglog(rad_arc, sersic_single(rad_arc, *optgal), label='Curve-fit', color='r')
plt.title('Radial-Flux-Profile')
```

```
plt.xlabel('Radius')
plt.ylabel('Flux Density')
plt.legend()
plt.show()
```

The * mark in front of the optgal variable indicates the program to take the values in sequence as it is stored in the variable.

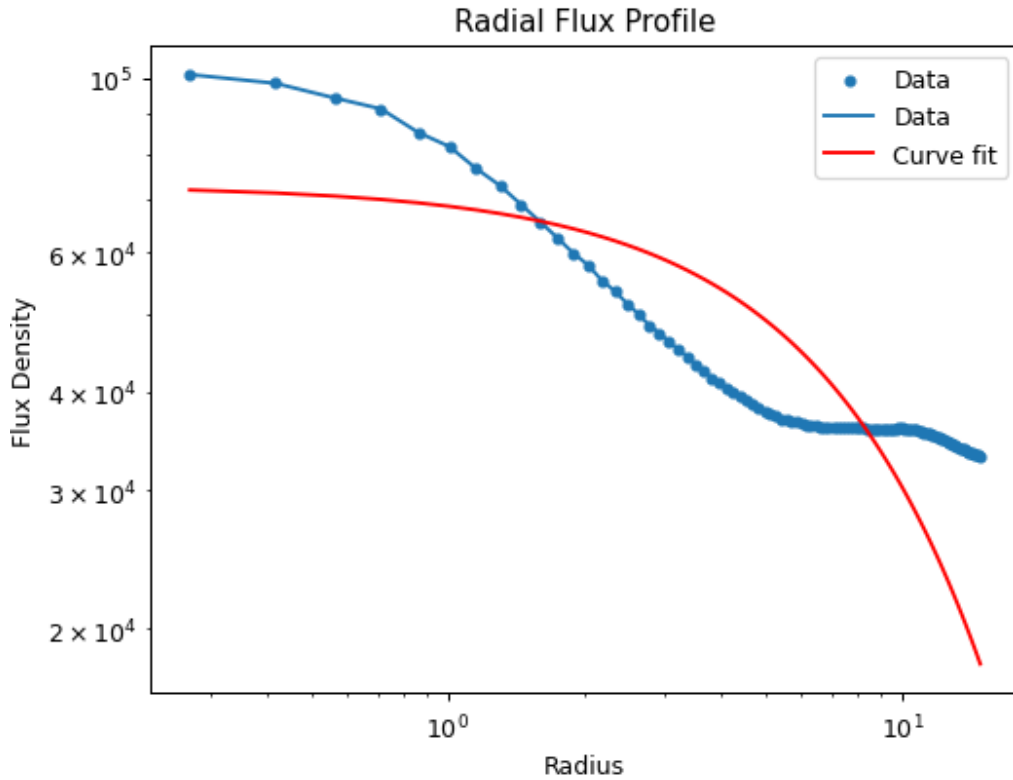


Figure 7.4: Single Sersic Fit

Listing 17: Best fit parameters I, R, n

```
Intensity : 18048.580133546362
Radius : 14.85
Sersic index : 0.858569696028098
```

I have got these values in Listing 17 which is a good fit for now. Sersic index value - 0.858 indicated that the galaxy is a spiral galaxy, which is true in this case because we are analysing for the galaxy NGC3982.

The micro-project meeting at 11:00AM at McKenzie. The galaxy group, all 3 of us have fit the single sersic fit. I am wondering whether I should be breaking my head on figuring out a way to subtract the star in front of the galaxies NGC7773 and NGC2865. The code is explained in the Desi Astro playlist on YouTube.

After the micro-project meeting I met up with Dr.Andreas Papageorgiou to discuss the code of the double sersic function. He patiently explained how the curve fit works and how the code of the double sersic fit works, that was available on Teams.

1. Curve fit takes the parameters that the user gives in and assumes those are the best values.
2. Next the function tries to measure the difference (in a sense) between the values guessed by the user and the actual values that the function is fitting against.
3. It takes a next value for the parameters, maybe in the positive direction or in the negative direction, finds the difference.
4. The latest difference is compared against the previous difference.
5. If the difference is small, the function assumes the parameters in that direction (can be positive or negative direction).

6. The bounds indicate the range of values the curve fit can assume to calculate the best fit.
7. The curve fit repeats this process a huge number of time until it gets close to the actual values it is trying to fit.
8. This fitting the best values also depends on the number of data points the user gives the function.

After explaining this to me, Dr. Andreas explained how the function is trying to fit the bulge and the disk in a similar manner in turn explaining how the double fit works.

7.7 Operational summary for week 11

- I have got the surface brightness right.
- I have figured out how the single fit works and got the values successfully.

7.8 Reflective statement for week 11

- Getting help from Dr. Andreas was really helpful. Which saved a lot of time and also helped me understand things that I would have not otherwise.
- Clarifying before doing anything will save time and help to go in the right direction of the project.

8 Week 12: 26 February 2024 – 01 March 2024

8.1 Week 12 objectives

- Get the double sersic fit

8.2 Monday 26 February 2024

I tried to fit the curve only to the bulge data points using the previously defined function (sersic single).

Listing 18: Sersic fit; Bulge (taken from the provided code on Teams and modified)

```
# Fit a single Sersic profile to our data
I_b = 800
R_b = 180
n_b = 2.9

#guess_1=sersic_single(rad_arc, I_b, R_b,n_b)

sel=rad_arc<6
optgal1, covgal1 = curve_fit(sersic_single,rad_arc[sel],flux_density[sel],
bounds=((0,rad_arc[0],0),(1e5,rad_arc[1e5]-1],10)), maxfev = 10000)
print(optgal1)
plt.scatter(rad_arc[sel],flux_density[sel],s=15, label='Data')
plt.loglog(rad_arc[sel],(flux_density[sel]))
plt.loglog(rad_arc,sersic_single(rad_arc,*optgal1), label='Curve-fit', color='r')
plt.title('Radial-Flux-Profile')
plt.xlabel('Radius')
plt.ylabel('Flux-Density')
plt.legend()
plt.show()
```

OUTPUT: [8.71859272e+03 1.48500000e+01 1.48225195e+00]

The code section in Listing 18, used to plot the curve fit only to the bulge section of the produced surface brightness profile. `sel=rad_arc<6` limits the data points that are less than 6 to be considered while the curve fit tries to find the best fit. Rest of the code lines are similar to discussed in the earlier days.

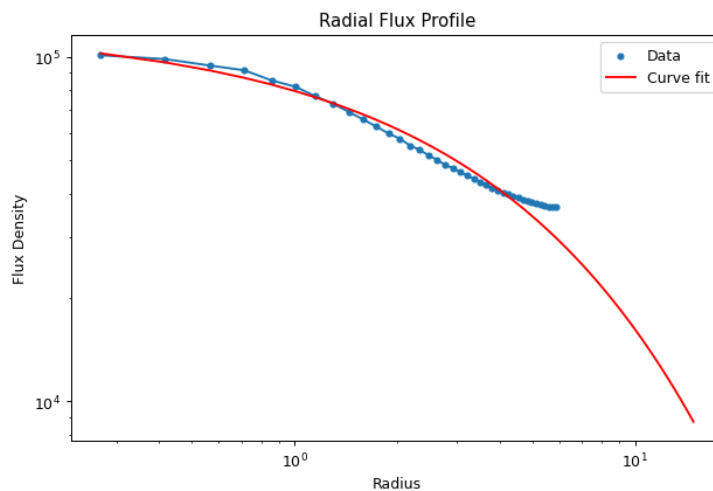


Figure 8.1: Sersic Fit - Bulge

I have tried to do the same with data points of the disk.

Listing 19: Sersic fit; Disk (taken from the provided code on Teams and modified)

```
# Fit a single Sersic profile to our data
```



```
I_b = 10400
R_b = 480
n_b = 0.8
```

```
guess_1=seraic_single(rad_arc , I_b , R_b,n_b)
```

```
sell=rad_arc>6
optgal2 , covgal2 = curve_fit(seraic_single ,rad_arc[sell],flux_density[sell] ,
bounds=((0,rad_arc[0],0),(1e5,rad_arc[len(rad_arc)-1],10)), maxfev = 10000)
```

OUTPUT: [3.26259314e+04 1.48500000e+01 1.70572525e-01]

Here, $sell=rad_arc>6$ considers data points only above the radius value 6, making the curve fit to be influenced by those particular data points. The I, R, n values 10400,480,0.8 respectively are guessed and input to the `seraic_single` function, which produces a curve with those values. I was told to manually figure out a set values that fits the curve best, just as an exercise to know how all these functions work together.

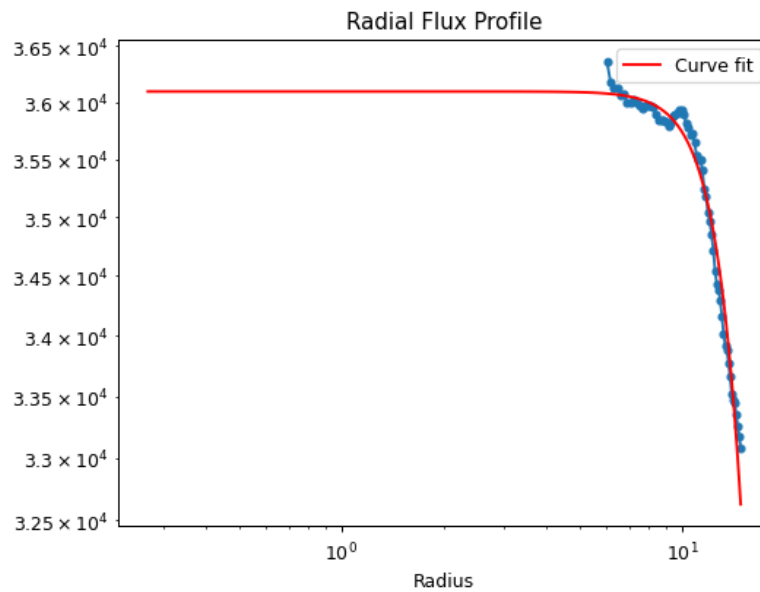


Figure 8.2: Sersic Fit - Disk

8.3 Tuesday 27 February 2024

No work done.

8.4 Wednesday 28 February 2024

I have plotted the previously obtained best fit values of bulge and disk onto the same graph to see how well those merge.

Listing 20: Plot the bulge and the disk curve fit together (taken from the provided code on Teams and modified)

```
plt.figure()
plt.scatter(rad_arc , flux_density , s=15, label='Data')
plt.loglog(rad_arc , seraic_single(rad_arc ,*optgal1) ,
label='Profile of bulge' , color='r')
plt.loglog(rad_arc , seraic_single(rad_arc ,*optgal2) ,
label='Profile of disk' , color='g')
plt.title('Radial Flux Profile')
plt.xlabel('Radius')
plt.ylabel('Flux Density')
plt.legend()
plt.show()
```

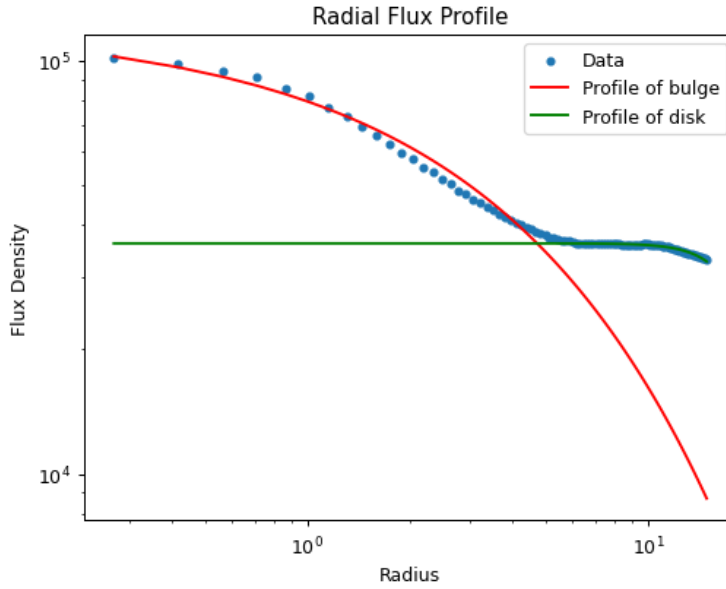


Figure 8.3: Sersic Fit - Bulge + Disk

Then I remembered Dipanjan asking me and Shyam, how did we know that the bulge curve would end at radius value 6 and disk would start there. It is not possible to guess by eye, where the curves classify into bulge and disk. So the plot Fig 8.3 does not make sense. It only shows that the fitting is working well. The fitting for the bulge and the disk should be done separately during the actual values are estimated in the Sersic1D[12] function. I cannot do them after I have got the values and just add the plots together. That is what Dipanjan explained it as. I looked at the code for the double sersic function which was posted on Teams.

Listing 21: Double Sersic function (taken from the provided code on Teams)

```
# Define Sersic function as a function
def sersic_double(R, i_b, r_b, n_b, i_d, r_d, n_d):
    return Sersic1D(i_b, r_b, n_b)(R) + Sersic1D(i_d, r_d, n_d)(R)
```

This piece of code defines the sersic double function that I need. The parameters are passed separately to each of the Sersic1D function to fit the bulge and the disk separately. I have had this wrong notion about fitting, imagining it as two lines being minimised like in a plot. Fitting just means that finding the best values that minimises the difference between the input parameters and the actual values. Plotting those values is a separate thing. Also another misconception I had is Sersic2D [12] is the double fit function. But in reality, Sersic2D is a 2D array modelling that is not related to the sersic fits I am dealing with. Sersic double fit just means that there are best values found separately for the bulge and the disk, that is all.

Listing 22: Curve fit for sersic double function (taken from the provided code on Teams and modified)

```
optgal3, covgal3 = curve_fit(sersic_double, rad_arc, flux_density,
    bounds=((0, rad_arc[0], 0, 0, rad_arc[0], 0), (1e5, rad_arc[len(rad_arc)-1], 10, 1e5,
    rad_arc[len(rad_arc)-1], 10)), maxfev=10000)
print("Double fit: ", optgal3)
```

```
# Define a function for the best fit curve for bulge
def galaxyprofile_bulge(r):
    return Sersic1D(*optgal3[0:3])(r)
```

```
# Define a function for the best fit curve for disk
def galaxyprofile_disc(r):
    return Sersic1D(*optgal3[3:6])(r)
```

Since I am trying to fit the curve for bulge and disk separately, I need to pass three range values each for bulge and disk separately. Taking a look at the curve fit function mentioned in Listing 22, it is clear that in the bounds two sets of 6 values are being passed to the function. I tried to plot all these values.

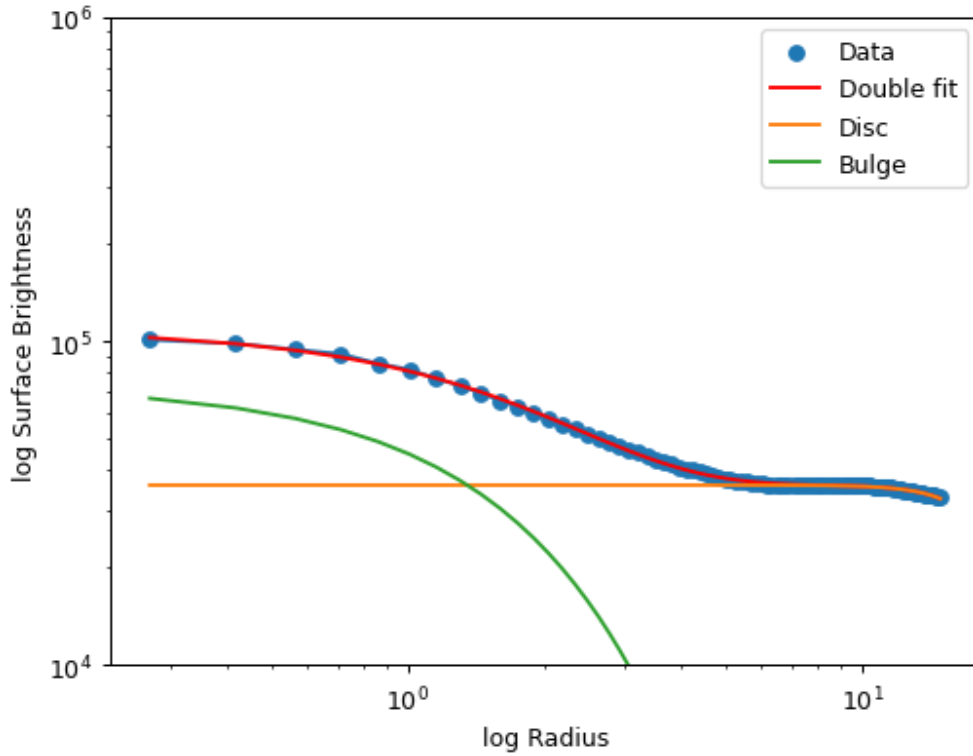


Figure 8.4: Double Sersic Fit

Listing 23: Output of double sersic fit

```
OUTPUT: Double fit : [2.06584849e+04 2.13461989e+00 7.90894814e-01
3.27016408e+04 1.48475671e+01 1.66687988e-01]
```

The bulge curve is not merging with the actual data points. That is one thing that I have to figure out. From the output the index value for the disk being measure is 0.1666 and for the bulge is 0.7908, which are good values but there is some mistake in the fitting.

8.5 Thursday 29 February 2024

I met up with Dipanjan in his office today to figure out what's going on in the plot. He said the single sersic fit looks fine even though it is not fitting very well. I am getting a value of 0.8585 as the sersic index (n) value. The double fit looks fine too but trying to plot just the bulge and just the disk curves on top of each other while plotting the double sersic fit makes the bulge curve fit to plot not on the data points. Dipanjan suggested me to play around with the bounds and he helped me with bounds to fit the single sersic fit better. The bounds that I was using initially were:

Listing 24: Bounds 1

```
bounds=((0,0,0,0,0,0),(1e4,1e4,10,1e4,1e4,10))
```

Which gave plots that were a bit off from the actual data.

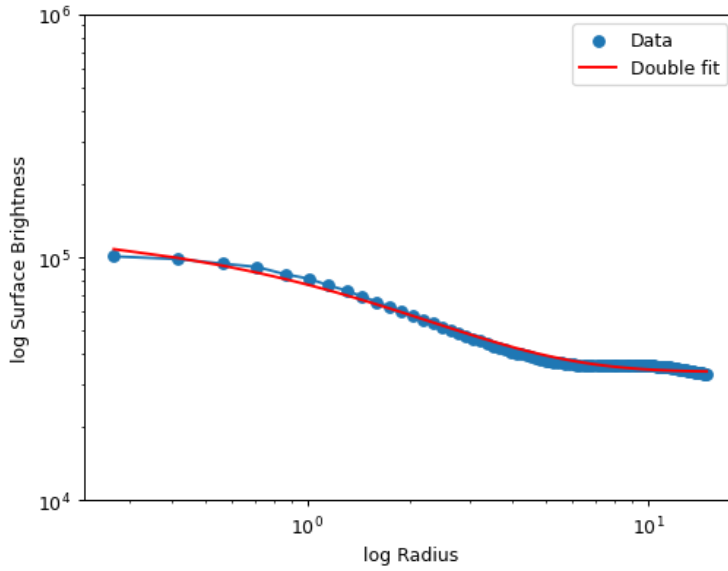


Figure 8.5: Initial bounds

Dipanjan helped me to redefine these bounds into values that the parameters can possibly assume within the actual data values.

Listing 25: Bounds 2

```
bounds=((0,rad_arc[0],0,0,rad_arc[0],0),
(1e5,rad_arc[len(rad_arc)-1],10,1e5,rad_arc[len(rad_arc)-1],10))
```

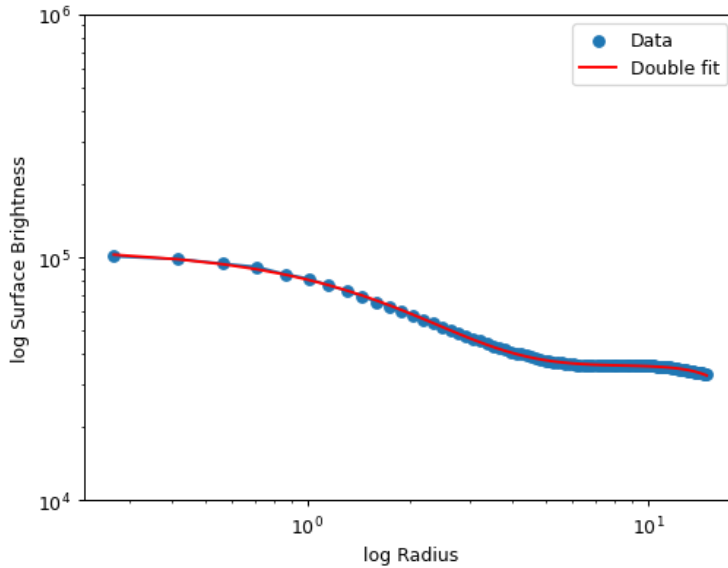


Figure 8.6: Updated bounds

Which gives a better fit. One thing to note is that the curve fitting should not be changing around just by changing the bounds. That is another thing I have to figure out.

8.6 Friday 01 March 2024

Micro-project meeting at 11:00AM at McKenzie. I have managed to get the double sersic fit which is matching the bulge and the disk well. When I try to plot the fit of the disk and the bulge separately, the plot is a mess as shown in the entries from the other day.

8.7 Operational summary for week 12

- Got the double sersic fit successfully.

8.8 Reflective statement for week 12

- Could have did the analysis bit quicker, I could have covered few other techniques to do the analysis and get the results.

9 Week 13: 04 March 2024 – 08 March 2024

9.1 Week 13 objectives

- Do any extra analysis possible.
- Get the research paper ready.

9.2 Monday 04 March 2024

I wanted to check whether background subtraction makes any difference in the plot. So I decided to use flux values that I got from using JS9 software during the initial weeks. I went through the same procedure as I have previously mentioned. The following are the results that I have got.

Listing 26: Single sersic fit results on JS9 data

Co—ordinates of highest flux value in cropped image of NGC3982: 105 124

Intensity : 15786.364074768067

Radius : 14.85

Sersic index : 0.8072511146691631

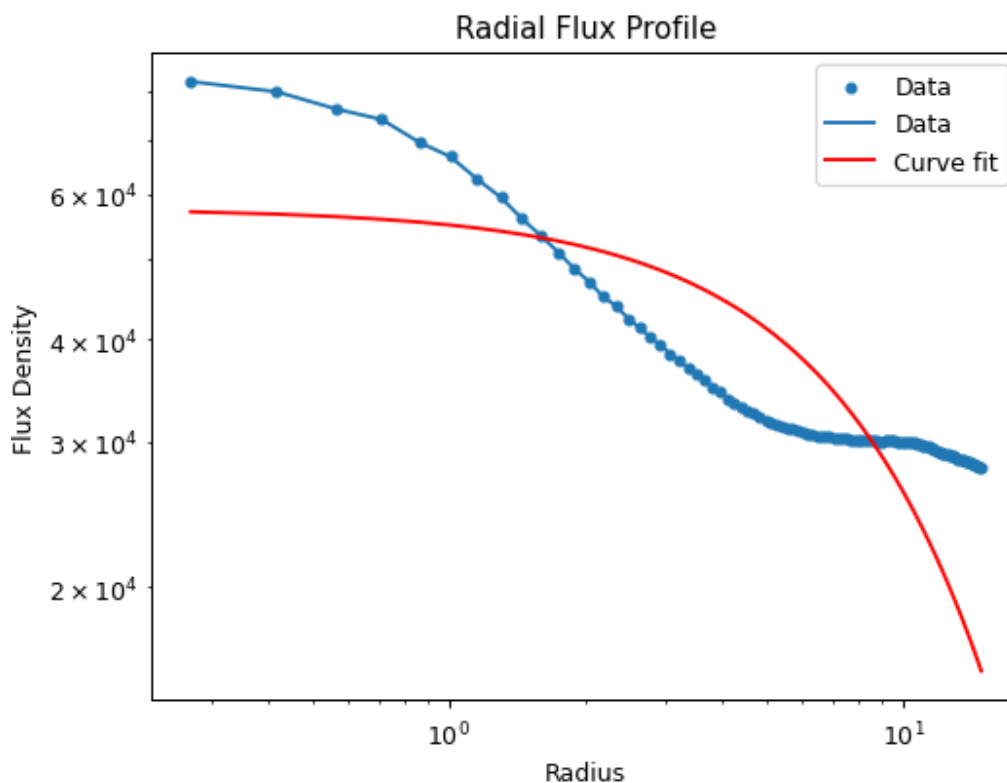


Figure 9.1: Single sersic fit results on JS9 data

The index value that I have got here is approximately near to the index value (0.8585) what I got when I used the data from drawing apertures in python without the background subtraction but there is a difference.

9.3 Tuesday 05 March 2024

I have had a lot of assignment due dates this week, because of which I could not work further on the micro-project.

- 9.4 Wednesday 06 March 2024
- 9.5 Thursday 07 March 2024
- 9.6 Friday 08 March 2024
- 9.7 Operational summary for week 13
- 9.8 Reflective statement for week 13

10 Reference list

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